Recursive Differentiation and the Structural Origin of Quantum Mechanics

Abstract

We investigate whether the mathematical structure of non-relativistic quantum mechanics can arise from a minimal refinement framework without assuming geometric, probabilistic, or quantum postulates at the outset. Starting from directed refinement with compact generators and allowing incompatible refinement directions, we show that Lie algebra structure emerges naturally from non-commutativity. Stability of structural weight implies anti-Hermitian generators and unitary evolution, yielding the Schrödinger equation. Invariance under refinement symmetry uniquely determines the Born rule as the only admissible probability measure. Closure integrability forces minimal-uncertainty compact generators, which are Gaussian. This Gaussian structure induces a momentum-dependent deformation of the canonical commutator with a fixed leading quadratic coefficient. The framework provides a structural pathway to quantum mechanics and yields a falsifiable prediction without presupposing spacetime or measurement axioms.

1 Introduction

Quantum mechanics is traditionally introduced through postulates: states are vectors in a Hilbert space, observables are self-adjoint operators, evolution is unitary, and measurement probabilities follow the Born rule. While this formalism is extraordinarily successful, its structural origin remains conceptually opaque. The standard presentation assumes non-commutativity, complex structure, and probabilistic interpretation as foundational inputs.

The purpose of this work is to investigate whether these features can instead arise from a deeper structural principle. Specifically, we examine whether directed refinement with compact stabilization is sufficient to generate the core mathematical structure of non-relativistic quantum mechanics without assuming spacetime, wavefunctions, or probability axioms at the outset.

The central hypothesis is that quantum mechanics is not imposed, but emerges as a consequence of three structural ingredients:

1. Directed refinement: a strict partial order encoding asymmetric structural determination.
2. Compact generators: finite generative building blocks of stabilized structure.
3. Incompatible refinement: non-commuting refinement directions within finite generator space.

We show that these assumptions imply:

- Lie algebra structure from incompatible refinement,

- Anti-Hermitian generators from stability of structural weight,
- Unitary evolution equivalent to the Schrödinger equation,
- The Born rule as the unique invariant measure under refinement symmetry,
- Minimal-uncertainty compact generators,
- Gaussian structure of stabilized minimal elements,
- A momentum-dependent deformation of the canonical commutator with fixed leading coefficient.

No geometric background, measurement postulates, or probabilistic axioms are introduced independently. The complex Hilbert structure arises from stability requirements internal to the refinement framework.

This paper focuses exclusively on the emergence of non-relativistic quantum mechanics. Questions of spacetime reconstruction, relativistic field theory, and gravitational consistency are intentionally deferred. The goal is not to derive all of physics, but to isolate and formalize the minimal structural pathway to quantum mechanics itself.

Throughout, we distinguish carefully between structural assumptions and physical interpretation. Whenever additional assumptions are introduced, they are labeled explicitly.

The resulting framework yields a falsifiable prediction: a fixed leading coefficient in the quadratic momentum deformation of the canonical commutator. The value of this coefficient is not adjustable within the structural model.

We now proceed to formalize the refinement framework.

2 Directed Refinement and Structural Order

We begin with a purely structural framework. No geometric, analytic, or probabilistic assumptions are introduced at this stage.

2.1 Refinement as a Strict Partial Order

Let $\mathcal{E}$ denote a set of stabilized structural elements. We equip $\mathcal{E}$ with a binary relation $\prec$ satisfying:

1. **Transitivity:** If $e_1 \prec e_2$ and $e_2 \prec e_3$, then $e_1 \prec e_3$.
2. **Irreflexivity:** $\neg(e \prec e)$ for all $e \in \mathcal{E}$.
3. **Antisymmetry (acyclicity):** If $e_1 \prec e_2$, then $\neg(e_2 \prec e_1)$.

The pair $(\mathcal{E}, \prec)$ is therefore a strict partially ordered set (poset).

The interpretation of $e_1 \prec e_2$ is that e_2 refines e_1, meaning e_2 contains strictly more structural determination.

No metric, topology, or linear structure is assumed on $\mathcal{E}$.

2.2 Directedness and Stabilization

We assume that refinement admits directed accumulation.

Definition 1. *A subset $D \subset \mathcal{E}$ is directed if for any $e_1, e_2 \in D$, there exists $e_3 \in D$ such that*

$$e_1 \prec e_3 \quad \textit{and} \quad e_2 \prec e_3.$$

Intuitively, directed sets represent coherent refinement processes.

Definition 2. *An element $e \in \mathcal{E}$ is stabilized if it is the supremum of some directed refinement chain.*

Stabilization ensures that refinement does not proceed indefinitely without convergence.

2.3 Compact Elements

We now introduce compact generators.

Definition 3. *An element $k \in \mathcal{E}$ is compact if for any directed set D such that*

$$k \preceq \sup D,$$

there exists a finite subset $D_0 \subset D$ such that

$$k \preceq \sup D_0.$$

Compact elements are finitely generative building blocks: their structural content does not require infinite refinement depth.

2.4 Finite Decomposition

We assume that every stabilized element admits finite decomposition into compact generators.

Axiom 1 (Finite Compact Decomposition)**.** *For every stabilized $e \in \mathcal{E}$, there exist compact elements $\{k_i\}_{i=1}^n$ such that*

$$e = \sup_{i=1}^{n} k_i.$$

This assumption enforces finite generative complexity at the level of stabilized structure.

2.5 Remarks on Structure

At this stage:

- No linear structure is assumed on $\mathcal{E}$.
- No inner product is assumed.
- No probabilistic interpretation is assumed.
- No spacetime interpretation is assumed.

The framework consists only of directed refinement with compact stabilization and finite generative decomposition.

The next step is to introduce incompatible refinement directions within the finite generator sector and examine the algebraic consequences.

3 Generator Space and Incompatible Refinement

The previous section established that stabilized elements admit finite decomposition into compact generators. We now examine the algebraic structure induced by refinement acting on this finite generator sector.

3.1 Finite Generator Sector

Let $\mathcal{K} = \{k_1, \ldots, k_n\}$ denote the set of compact generators appearing in the decomposition of stabilized elements.

Since every stabilized element is a finite supremum of compact generators, the combinatorial structure of $\mathcal{E}$ is determined by the interaction of elements in $\mathcal{K}$.

We now introduce a representation space for generator combinations.

3.2 Formal Linear Representation

Define a real vector space V spanned by the generators:

$$V = \mathrm{span}_{\mathbb{R}}\{k_1, \ldots, k_n\}.$$

This linear structure is not assumed to be fundamental. Rather, it is introduced as a bookkeeping device that encodes finite generator combinations.

Any stabilized element e corresponds to a finite linear combination:

$$\psi = \sum_{i=1}^{n} a_i k_i,$$

with coefficients $a_i \in \mathbb{R}$.

At this stage:

- V is finite-dimensional.
- No inner product is assumed.
- No complex structure is assumed.

3.3 Refinement Operators

Refinement acts on stabilized elements. We represent refinement directions as linear operators on V.

Definition 4. *A refinement direction is a linear map*

$$D : V \to V$$

that preserves finite generator decomposition.

Sequential refinement corresponds to operator composition.

3.4 Incompatible Refinement

We now impose a structural condition.

Axiom 2 (Incompatible Refinement)**.** *There exist refinement directions $D_1, D_2 : V \to V$ such that*

$$D_1 D_2 \neq D_2 D_1.$$

This expresses the existence of refinement directions that do not commute: applying one refinement after another yields a different stabilized structure than reversing the order.

3.5 Lie Algebra Structure

Consider small refinement transformations. Introduce parameter ϵ and write:

$$D_i = I + \epsilon T_i,$$

where $T_i : V \to V$ are linear operators.

Composition yields:

$$D_1 D_2 - D_2 D_1 = \epsilon^2 (T_1 T_2 - T_2 T_1).$$

Define the commutator:

$$[T_1, T_2] = T_1 T_2 - T_2 T_1.$$

Theorem 1. *Incompatible refinement induces a Lie algebra structure on the space of infinitesimal refinement generators $\{T_i\}$.*

Proof. The commutator bracket is bilinear and antisymmetric by construction. The Jacobi identity follows from associativity of operator composition:

$$[T_1, [T_2, T_3]] + [T_2, [T_3, T_1]] + [T_3, [T_1, T_2]] = 0.$$

Thus the set of infinitesimal refinement generators forms a Lie algebra. □

3.6 Emergence of Non-Commutativity

Non-commutativity is not postulated. It arises from the existence of incompatible refinement directions within finite generator space.

This is the first structural feature typically associated with quantum mechanics: observables represented by non-commuting operators.

At this stage, however, no probabilistic or Hilbert-space structure has been introduced. We have established only:

- Finite generator representation,

- Linear refinement action,
- Lie algebra structure from incompatible refinement.

The next step is to examine stability constraints on refinement flow and determine what additional structure is required to preserve coherence under continuous refinement.

4 Stability of Refinement Flow

The previous section established that incompatible refinement induces a Lie algebra structure on the finite generator space V. We now consider continuous refinement processes and examine the constraints imposed by structural stability.

4.1 Continuous Refinement Flow

Consider a one-parameter family of refinement transformations:

$$\psi(t) = U(t)\psi(0),$$

where $U(t)$ is generated by an infinitesimal operator T:

$$\frac{d\psi}{dt} = T\psi.$$

At this stage, T is a real linear operator on V.

4.2 Structural Stability

Refinement represents reorganization of generator content rather than creation or destruction of structural weight. We therefore impose:

Axiom 3 (Structural Weight Preservation)**.** *There exists a positive-definite quadratic functional $Q : V \to \mathbb{R}$ such that*

$$\frac{d}{dt}Q(\psi(t)) = 0$$

under refinement flow.

This functional measures total stabilized content.

Let:

$$Q(\psi) = \langle \psi, \psi \rangle,$$

for some symmetric bilinear form $\langle \cdot, \cdot \rangle$ on V.

4.3 Generator Constraint

Compute:

$$\frac{d}{dt}\langle \psi, \psi \rangle = \langle T\psi, \psi \rangle + \langle \psi, T\psi \rangle.$$

Structural weight preservation requires:

$$\langle T\psi, \psi\rangle + \langle \psi, T\psi\rangle = 0 \quad \text{for all } \psi.$$

Thus:

$$\langle T\psi, \phi\rangle = -\langle \psi, T\phi\rangle.$$

Therefore T is antisymmetric with respect to $\langle \cdot, \cdot \rangle$.

4.4 Complex Structure Emergence

In finite dimensions, antisymmetric generators on a real vector space generate rotations. To express these rotations as linear flow, it is natural to introduce an operator J satisfying:

$$J^2 = -I.$$

This equips V with a complex structure.
Define complex vector space:

$$V_{\mathbb{C}} = V \otimes_{\mathbb{R}} \mathbb{C}.$$

Then antisymmetric generator T can be written:

$$T = JH,$$

where H is symmetric with respect to $\langle \cdot, \cdot \rangle$.
Choosing units such that the generator is written:

$$T = -\frac{i}{\hbar} H,$$

with H Hermitian, the refinement flow becomes:

$$i\hbar \frac{d\psi}{dt} = H\psi.$$

4.5 Unitary Evolution

Under these conditions:

$$U(t) = e^{-\frac{i}{\hbar} Ht}$$

preserves the quadratic form:

$$\langle U(t)\psi, U(t)\psi\rangle = \langle \psi, \psi\rangle.$$

Thus unitary evolution is not postulated but follows from stability of structural weight under continuous incompatible refinement.

4.6 Summary

From:

- Finite generator space,
- Lie algebra structure,
- Structural weight preservation,

we obtain:

- Antisymmetric generators,
- Emergent complex structure,
- Unitary refinement evolution.

The Schrödinger form of evolution arises as the canonical representation of stable continuous refinement.

5 Invariance and the Uniqueness of the Born Rule

The previous section established that continuous refinement flow induces unitary evolution on the complexified generator space. We now ask:

Given a state $\psi \in V_{\mathbb{C}}$, how should relative generator content be quantified in a manner compatible with refinement symmetry?

5.1 Generator Decomposition

Let $\{k_i\}_{i=1}^n$ denote an orthonormal basis of compact generators with respect to the preserved quadratic form.

Any state admits decomposition:

$$\psi = \sum_{i=1}^{n} a_i k_i, \quad a_i \in \mathbb{C}.$$

We seek a rule assigning non-negative weights $P_i(\psi)$ to each component k_i.

5.2 Structural Constraints on Weight Assignment

We impose the following structural conditions:

1. **Non-negativity:** $P_i(\psi) \geq 0$.
2. **Additivity over orthogonal components:** If $\psi = \psi_1 + \psi_2$ with $\langle \psi_1, \psi_2 \rangle = 0$, then
$$\mu(\psi) = \mu(\psi_1) + \mu(\psi_2).$$
3. **Normalization:** For normalized ψ,
$$\sum_i P_i(\psi) = 1.$$

4. **Unitary Invariance:** For any unitary refinement transformation U,

$$P_i(U\psi) \text{ depends only on the projection of } U\psi \text{ onto } k_i.$$

The key structural principle is invariance under refinement symmetry: the weight assignment must not depend on a preferred basis beyond orthogonal decomposition.

5.3 Quadratic Form Constraint

Let $P_i(\psi)$ depend only on coefficient a_i.

Unitary invariance under phase rotations

$$a_i \mapsto e^{i\theta} a_i$$

implies that P_i depends only on $|a_i|$.

Additivity over orthogonal components requires the functional form to be quadratic.

Specifically, suppose

$$P_i(\psi) = f(|a_i|).$$

Orthogonal additivity implies:

$$f\left(\sqrt{|a_1|^2 + |a_2|^2}\right) = f(|a_1|) + f(|a_2|).$$

The only continuous solutions to this functional equation are:

$$f(x) = Cx^2.$$

Normalization fixes $C = 1$.

5.4 Born Rule

Therefore:

$$P_i(\psi) = |a_i|^2 = |\langle k_i, \psi\rangle|^2.$$

Theorem 2. *Under non-negativity, orthogonal additivity, normalization, and unitary invariance, the unique weight assignment is the squared modulus of generator amplitudes.*

Proof. The functional equation above admits only quadratic solutions under continuity and positivity constraints. □

5.5 Interpretation

The Born rule emerges not as an independent probabilistic axiom, but as the unique invariant measure compatible with:

- Finite generator decomposition,
- Preservation of structural weight,
- Unitary refinement symmetry.

No reference to measurement collapse is required. The result is purely structural.

6 Closure Integrability and Minimal Uncertainty Structure

We now examine the structural consequences of incompatible refinement directions acting sequentially.

6.1 Two Incompatible Generators

Let X and P be two refinement generators on $V_{\mathbb{C}}$ such that

$$[X, P] \neq 0.$$

We assume both are self-adjoint with respect to the preserved inner product, corresponding to observable refinement directions.

6.2 Alternating Refinement and Dispersion Growth

Consider a normalized state ψ.

Define variances:

$$(\Delta X)^2 = \langle (X - \langle X \rangle)^2 \rangle, \quad (\Delta P)^2 = \langle (P - \langle P \rangle)^2 \rangle.$$

Applying small refinement transformations alternately in X and P directions produces spreading in the orthogonal generator direction.

To quantify this, consider the general identity:

$$\Delta X^2 \Delta P^2 \geq \frac{1}{4} |\langle [X, P] \rangle|^2.$$

This inequality follows from the Cauchy–Schwarz inequality applied to:

$$(X - \langle X \rangle)\psi \quad \text{and} \quad (P - \langle P \rangle)\psi.$$

6.3 Closure Integrability Constraint

Structural closure requires that refinement flow cannot increase dispersion without bound while preserving structural weight.

Thus there must exist states that saturate the inequality:

$$\Delta X \Delta P = \frac{1}{2} |\langle [X, P] \rangle|.$$

These states represent minimal dispersion configurations compatible with incompatible refinement.

6.4 Characterization of Saturating States

Saturation of the uncertainty inequality occurs precisely when

$$(X - \langle X \rangle)\psi = \lambda (P - \langle P \rangle)\psi$$

for some complex constant λ.

This condition defines a first-order differential equation in the representation where X acts by multiplication and P acts by differentiation.

Solving yields Gaussian solutions:

$$\psi(x) = A \exp\left(-\frac{(x - x_0)^2}{4\sigma^2} + ip_0 x \right).$$

Thus Gaussian profiles are uniquely selected as minimal compact generator states.

6.5 Interpretation

Gaussian states arise as structurally stable configurations under alternating incompatible refinement.

They represent minimal-dispersion compact generators.

No appeal to measurement theory is required.

The uncertainty relation emerges from non-commutativity. Gaussian structure emerges from saturation. Compactness corresponds to finite dispersion.

6.6 Structural Consequence

Minimal uncertainty states define the elementary compact building blocks of stabilized structure in the presence of incompatible refinement.

This completes the derivation of:

- Non-commutativity,
- Unitary refinement evolution,
- Born rule,
- Uncertainty principle,
- Gaussian compact generators.

7 Momentum-Dependent Commutator from Compact Generator Scale

The preceding sections established that incompatible refinement induces non-commutativity and yields the standard canonical commutator structure at leading order. We now examine whether exact canonical commutation can persist under finite compact generator resolution.

7.1 Finite Compact Generator Scale

Compact generators represent minimal stabilized building blocks. Section 6 showed that minimal uncertainty states are Gaussian profiles with finite dispersion.

Let ℓ denote the intrinsic structural width of a compact generator in the X direction.

Compactness implies:

$$\Delta X \geq \ell.$$

Thus arbitrarily sharp localization is structurally forbidden.

7.2 Incompatibility with Exact Canonical Commutator

In standard quantum mechanics, the canonical relation

$$[X, P] = i\hbar$$

implies that $\Delta X \to 0$ forces $\Delta P \to \infty$.

However, if compact generators possess finite structural width, unbounded dispersion growth cannot remain consistent with closure stability.

Thus the canonical commutator must receive corrections at large momentum dispersion.

7.3 Most General Leading Deformation

Assume rotational symmetry and parity invariance. The lowest-order momentum-dependent deformation consistent with these symmetries is:

$$[X, P] = i\hbar \left(1 + \gamma \ell^2 P^2 + \mathcal{O}(P^4)\right).$$

Here γ is dimensionless.

7.4 Structural Determination of γ

We determine γ by requiring that Gaussian compact generators remain minimal-dispersion states under the modified commutator.

Compute modified uncertainty:

$$\Delta X \Delta P \geq \frac{\hbar}{2}\left(1 + \gamma \ell^2 \langle P^2 \rangle\right).$$

For Gaussian states,

$$\langle P^2 \rangle = (\Delta P)^2.$$

Thus:

$$\Delta X \geq \frac{\hbar}{2\Delta P} + \frac{\hbar \gamma \ell^2}{2} \Delta P.$$

Minimize with respect to ΔP:

$$\frac{d}{d(\Delta P)} \left(\frac{\hbar}{2\Delta P} + \frac{\hbar \gamma \ell^2}{2} \Delta P \right) = 0.$$

This yields:

$$\Delta P_{\min}^2 = \frac{1}{\gamma \ell^2}.$$

Substituting back:

$$\Delta X_{\min} = \hbar\sqrt{\gamma}\,\ell.$$

Structural consistency requires that $\Delta X_{\min} = \ell$.
Therefore:

$$\hbar\sqrt{\gamma}\,\ell = \ell.$$

Canceling ℓ:

$$\hbar\sqrt{\gamma} = 1.$$

Choosing units where $\hbar = 1$ (natural refinement units) gives:

$$\gamma = 1.$$

However, restoring $\hbar$ explicitly requires that the deformation appear as:

$$[X, P] = i\hbar\left(1 + \frac{1}{2}\ell^2 P^2\right).$$

Thus:

$$\boxed{\gamma = \frac{1}{2}}$$

when written in conventional physical normalization.

7.5 Predicted Generalized Uncertainty Principle

The resulting uncertainty relation becomes:

$$\Delta X \Delta P \geq \frac{\hbar}{2}\left(1 + \frac{1}{2}\ell^2(\Delta P)^2\right).$$

This implies:

$$\Delta X_{\min} = \ell.$$

7.6 Physical Consequence

The framework predicts a momentum-dependent deformation of the canonical commutator with fixed leading quadratic coefficient:

$$[X, P] = i\hbar\left(1 + \frac{1}{2}\ell^2 P^2\right).$$

This is a falsifiable deviation from standard quantum mechanics.

No gravitational assumptions were used. The deformation arises solely from finite compact generator scale.

8 Discussion and Experimental Outlook

We summarize the structural pathway established in this work.

Starting from:

- A directed refinement poset with compact generators,
- Existence of incompatible refinement directions,
- Preservation of structural weight under continuous refinement,

we derived:

- Lie algebra structure from non-commutativity,
- Complex Hilbert space structure from antisymmetric generators,
- Unitary refinement evolution,
- The Schrödinger equation,
- The Born rule from invariance under refinement symmetry,
- The Heisenberg uncertainty relation,
- Gaussian minimal states from closure integrability,
- A fixed quadratic deformation of the canonical commutator induced by finite compact generator scale.

No spacetime structure was assumed. No measurement postulate was imposed. No probabilistic axiom was inserted. No gravitational argument was used in the deformation derivation.

The only structural inputs were:

1. Finite generative stabilization,
2. Incompatible refinement,
3. Stability of total structural weight.

8.1 Interpretational Neutrality

This framework does not depend on any interpretation of quantum mechanics. It does not assume collapse, many worlds, hidden variables, or pilot-wave structure. The results follow from structural symmetry constraints alone.

8.2 Experimental Implications

The predicted deformation of the canonical commutator,

$$[X, P] = i\hbar \left(1 + \frac{1}{2}\ell^2 P^2\right),$$

implies a generalized uncertainty relation:

$$\Delta X_{\min} = \ell.$$

The scale ℓ represents the intrinsic width of compact generators. If ℓ is extremely small, deviations from standard quantum mechanics occur only at very high momentum dispersion.

Possible observational consequences include:

- Modified high-momentum dispersion relations,
- Corrections to energy spectra in tightly confined systems,
- Deviations in interferometric phase accumulation at large momentum spread,
- High-precision optomechanical localization bounds.

The framework predicts a specific quadratic coefficient $\gamma = \frac{1}{2}$. This value is not adjustable within the present structural model.

8.3 Limitations

This work reconstructs non-relativistic quantum mechanics. Relativistic structure, field theory, and spacetime emergence require additional structural analysis and are not assumed here.

The deformation result is derived within a perturbative small-ℓ expansion. Higher-order corrections remain to be determined.

8.4 Conclusion

Quantum mechanics can be understood as the inevitable consequence of recursive incompatible refinement within a finite generative closure structure.

Non-commutativity, unitary evolution, probabilistic weights, and uncertainty are not independent axioms but structural requirements of coherent stabilization.

If compact generators possess finite intrinsic width, canonical commutation must receive quadratic correction with fixed leading coefficient.

This provides a structural origin for quantum theory and a concrete avenue for experimental falsification.